An Unstable Variant of the Morava Change of Rings Theorem for $K(n)$ Theory

Lean 4 Blueprint

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Chapter 1

Introduction

The aim of this work is twofold: to give a foundational account of comonadic techniques in unstable homotopy theory, and to apply this foundation to prove an unstable variant of the Morava change of rings theorem from stable homotopy theory [4, 5], completely formalized in Lean 4 [3, 6].

Comonads have traditionally been used to define resolving objects, and hence derived functors, in both abelian and non-abelian settings [1]. In work of Bousfield [2], a comonad G defines a model structure on the category of cosimplicial objects over the category $\mathcal{C}^G$ of coalgebras over G, in which the traditional cobar resolutions compute derived mapping spaces and Ext groups.

Under suitable hypotheses, a morphism of comonads induces a functor on the categories of cosimplicial coalgebras that preserves resolving objects. This is the content of our *Generalized Shapiro's Lemma for Comonads* (Proposition 2.15). We then define an unstable version U_Σ of the Morava stabilizer algebra $\Sigma(n)$ and use the generalized Shapiro's lemma to prove that the unstable change of rings isomorphism holds in connectivity $k \geq 2p^n + 1$ (Theorem 3.9).

Chapter 2

The Theory of Comonads

2.1 Comonads, Coalgebras, and the Cokleisli Category

Definition 2.1 (Cokleisli-to-Coalgebra Functor)**.** For a comonad G on a category $\mathcal{C}$, the functor $i : \mathcal{C}_G \to \mathcal{C}^G$ sends $X \mapsto (GX, \Delta_X)$ and a Kleisli morphism $f : GX \to Y$ to $G(f) \circ \Delta_X : GX \to GY$.

Definition 2.2 (Coalgebra-to-Cokleisli Functor)**.** For a comonad G on $\mathcal{C}$, the functor $j : \mathcal{C}^G \to \mathcal{C}_G$ sends a G-coalgebra (X, ψ_X) to X and a coalgebra morphism $f : X \to Y$ to $f \circ \epsilon_X : GX \to Y$.

Proposition 2.3 (Cokleisli–Coalgebra Comparison)**.** *The functor $i : \mathcal{C}_G \to \mathcal{C}^G$ is fully faithful (exhibiting $\mathcal{C}_G$ as the full subcategory of cofree coalgebras in $\mathcal{C}^G$), and $j : \mathcal{C}^G \to \mathcal{C}_G$ provides the comparison functor in the opposite direction.*

Proof. Verified via the hom-set equivalence $\mathcal{C}(GX, Y) \simeq \mathcal{C}^G((X, \psi_X), (GY, \Delta_Y))$ sending $f \mapsto G(f) \circ \Delta_X \circ \psi_X$ with inverse $g \mapsto \epsilon_Y \circ g$. □

2.2 Morphisms of Comonads and Induced Functors

Definition 2.4 (Functor of Comonads)**.** A functor of comonads $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$ consists of a functor $F : \mathcal{C} \to \mathcal{D}$ and a natural transformation $\alpha : F \circ G \to H \circ F$ satisfying $\Delta^H_{FX} \circ \alpha_X = H(\alpha_X) \circ \alpha_{GX} \circ F(\Delta^G_X)$ and $\epsilon^H_{FX} \circ \alpha_X = F(\epsilon^G_X)$.

Proposition 2.5 (Coalgebra Pushforward)**.** *A functor of comonads* $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$ *induces a pushforward functor* $\alpha_* : \mathcal{C}^G \to \mathcal{D}^H$ *given on objects by* $\alpha_*(X, \psi_X) = (FX, \alpha_X \circ F(\psi_X))$*, commuting strictly with the forgetful functors.*

Proof. Coassociativity and counitality of $\alpha_X \circ F(\psi_X)$ follow directly from the coalgebra axioms for (X, ψ_X), naturality of α, and Definition 2.4. □

Definition 2.6 (Opfunctor of Comonads)**.** An opfunctor of comonads $(U, \beta) : (\mathcal{D}, H) \to (\mathcal{C}, G)$ consists of a functor $U : \mathcal{D} \to \mathcal{C}$ and a natural transformation $\beta : G \circ U \to U \circ H$ satisfying $U(\Delta^H_Y) \circ \beta_Y = \beta_{HY} \circ G(\beta_Y) \circ \Delta^G_{UY}$ and $U(\epsilon^H_Y) \circ \beta_Y = \epsilon^G_{UY}$.

Proposition 2.7 (Cokleisli Pushforward)**.** *An opfunctor of comonads* $(U, \beta) : (\mathcal{D}, H) \to (\mathcal{C}, G)$ *induces a functor* $\beta_* : \mathcal{D}_H \to \mathcal{C}_G$ *given by* $\beta_*(Y) = UY$ *and* $\beta_*(f : HY \to Y') = U(f) \circ \beta_Y$*.*

Proof. Preservation of identities and composition follows from the two opfunctor axioms of Definition 2.6. □

Proposition 2.8 (Coalgebra Pullback via Equalizers)**.** *Assume* $\mathcal{C}^G$ *has equalizers. Any opfunctor of comonads* $(U, \beta) : (\mathcal{D}, H) \to (\mathcal{C}, G)$ *induces a pullback functor* $\alpha^* : \mathcal{D}^H \to \mathcal{C}^G$ *defined on* (Y, ψ_Y) *as the equalizer in* $\mathcal{C}^G$ *of* $\mathrm{cofree}_G(U\psi_Y)$ *and* $G(\beta_Y) \circ \Delta^G_{UY} : \mathrm{cofree}_G(UY) \rightrightarrows \mathrm{cofree}_G(UHY)$*.*

Proof. Constructed via the limit equalizer of the parallel pair of cofree G-coalgebra morphisms. □

2.3 The Mate Correspondence and Lifted Adjunctions

Proposition 2.9 (Fundamental Mate Relations)**.** *Given* $F \dashv U$ *with unit* $\sigma : 1_{\mathcal{C}} \to UF$ *and counit* $\tau : FU \to 1_{\mathcal{D}}$*, and a comonad functor* $(F, \alpha) : (\mathcal{C}, G) \to (\mathcal{D}, H)$*, the mate transformation* $\beta_Y := U(H(\tau_Y) \circ \alpha_{UY}) \circ \sigma_{GUY}$ *satisfies*

$$\tau_{HY} \circ F(\beta_Y) = H(\tau_Y) \circ \alpha_{UY} \quad \text{and} \quad \beta_{FX} \circ G(\sigma_X) = U(\alpha_X) \circ \sigma_{GX}.$$

Proof. Proved using naturality of τ, σ, α and the triangle identities for $F \dashv U$. □

Proposition 2.10 (Mate Opfunctor of Comonads)**.** *Under the hypotheses of Proposition 2.9,* (U, β) *is an opfunctor of comonads* $(\mathcal{D}, H) \to (\mathcal{C}, G)$*.*

Proof. Verified by passing to left-adjoint transposes via Proposition 2.9 and applying the comonad functor laws of (F, α). □

Proposition 2.11 (Lifted Adjunction on Coalgebras)**.** *When $\mathcal{C}^G$ has equalizers, the pushforward $\alpha_* : \mathcal{C}^G \to \mathcal{D}^H$ is left adjoint to the pullback $\alpha^* : \mathcal{D}^H \to \mathcal{C}^G$.*

Proof. Constructed by lifting the unit σ_X into the equalizer $\alpha^*\alpha_* X$ via $G(\sigma_X) \circ \psi_X$, lifting the counit τ_Y via $\tau_Y \circ F(\epsilon^G_{UY} \circ \iota_Y)$, and verifying both triangle identities in $\mathcal{D}^H$ and $\mathcal{C}^G$. □

2.4 The Cobar Resolution and Left Contractibility

Definition 2.12 (Induced Monad and Cosimplicial Cobar Complex)**.** For a comonad G on $\mathcal{C}$, the adjunction $U_G \dashv \mathrm{cofree}_G$ induces a monad (T, μ, η) on $\mathcal{C}^G$ with $T(X, \psi_X) = (GX, \Delta_X)$, $\eta_X = \psi_X$, and $\mu_X = G(\epsilon_X)$. The cosimplicial cobar complex $C^\bullet_G(X)$ has $C^n_G(X) = T^{n+1}(X)$ with coaugmentation $d^0_{-1} = \eta_X$, cofaces $d^i_n : T^{n+1}X \to T^{n+2}X$, and codegeneracies $s^j_n : T^{n+2}X \to T^{n+1}X$.

Proposition 2.13 (Cosimplicial Identities and Left Contractibility into Cofree Coalgebras)**.** *The cobar complex $C^\bullet_G(X)$ satisfies all five cosimplicial identities, and for every $J \in \mathcal{C}$ the augmented simplicial hom-tower $\mathrm{Hom}_{\mathcal{C}^G}(C^\bullet_G(X), G(J)) \to \mathrm{Hom}_{\mathcal{C}^G}(X, G(J))$ admits a left contraction given by $f \mapsto G(f) \circ \Delta$.*

Proof. All five cosimplicial identities follow by induction on cosimplicial degree from the monad laws of T. The left contraction identities follow from coassociativity and counitality of G. □

2.5 A Generalized Shapiro's Lemma for Comonads

Proposition 2.14 (Shapiro Naturality Triangle)**.** *For any $f : M \to \alpha^* N$ in $\mathcal{C}^G$, the lifted adjunction $\tilde{\gamma} : \mathrm{Hom}_{\mathcal{C}^G}(M, \alpha^* -) \simeq \mathrm{Hom}_{\mathcal{D}^H}(\alpha_* M, -)$ commutes with the cobar coaugmentation $\psi_N : N \to H(N)$:*

$$\tilde{\gamma}_{M,H(N)}(f \circ_{post} \alpha^*(\psi_N)) = \tilde{\gamma}_{M,N}(f) \circ \psi_N.$$

Proof. Proved via naturality of the lifted unit and the adjunction hom-equivalence. □

Proposition 2.15 (Generalized Shapiro's Lemma for Comonads)**.** *Assume there is a functor $W : \mathcal{D} \to \mathcal{C}$ and a natural isomorphism $U_G \circ \alpha^* \cong W \circ U_H$. Then:*

1. *For every $N \in \mathcal{D}^H$ and $J \in \mathcal{C}$, the hom-tower $\mathrm{Hom}_{\mathcal{C}^G}(\alpha^*(C_H^\bullet(N)), G(J)) \to \mathrm{Hom}_{\mathcal{C}^G}(\alpha^* N, G(J))$ is left-contractible, so $\alpha^* N \to \alpha^*(C_H^\bullet(N))$ is a Bousfield weak G-resolution.*

2. *For every $M \in \mathcal{C}^G$ and $N \in \mathcal{D}^H$, there is an isomorphism of coaugmented cosimplicial hom-towers $\mathrm{Hom}_{\mathcal{C}^G}(M, \alpha^*(C_H^\bullet(N))) \cong \mathrm{Hom}_{\mathcal{D}^H}(\alpha_* M, C_H^\bullet(N))$, inducing $\mathrm{Ext}_G(M, \alpha^* N) \cong \mathrm{Ext}_H(\alpha_* M, N)$.*

Proof. Part (1) transports the underlying left contraction $W(\epsilon^H)^*$ across the adjunction $U_G \dashv \mathrm{cofree}_G$ and the isomorphism $U_G \circ \alpha^* \cong W \circ U_H$. Part (2) applies the lifted adjunction levelwise across the cobar complex. □

Chapter 3

An Unstable Change of Rings

3.1 The Stable Case and Canonical Coalgebra Equalizer

Proposition 3.1 (Canonical Equalizer Presentation of Coalgebras)**.** *For any comonad G on $\mathcal{C}$ and any G-coalgebra $X = (X, \psi_X)$, the coaction $\psi_X : X \to \mathrm{cofree}_G(X)$ exhibits X as the equalizer in $\mathcal{C}^G$ of $\mathrm{cofree}_G(\psi_X)$ and $\Delta_X : \mathrm{cofree}_G(X) \rightrightarrows \mathrm{cofree}_G(GX)$.*

Proof. Any cone $k : W \to \mathrm{cofree}_G(X)$ equalizing $\mathrm{cofree}_G(\psi_X)$ and Δ_X factors uniquely through X via $\epsilon_X \circ k$. □

Theorem 3.2 (Stable Change of Rings)**.** *When the stable cotensor satisfies $U_G \circ \pi^* \cong (B(n)_* \otimes_{K(n)_*} -) \circ U_H$, Proposition 2.15 yields the isomorphism of cobar hom-towers $\mathrm{Hom}_{\mathcal{C}^G}(M, \pi^* C_H^\bullet(N)) \cong \mathrm{Hom}_{\mathcal{D}^H}(\pi_* M, C_H^\bullet(N))$.*

Proof. Direct specialization of Proposition 2.15. □

3.2 The Unstable Case

Theorem 3.3 (Unstable Sandwich Comonad U_Σ)**.** *For any comonad U_Γ on $\mathcal{C}$ and any adjunction $F \dashv U : \mathcal{D} \to \mathcal{C}$ with unit η and counit ε, the endofunctor $U_\Sigma := F \circ U_\Gamma \circ U : \mathcal{D} \to \mathcal{D}$ is a comonad with counit $\epsilon_N^\Sigma = \varepsilon_N \circ F(\epsilon_{UN}^\Gamma)$ and comultiplication $\Delta_N^\Sigma = F(U_\Gamma(\eta_{U_\Gamma UN})) \circ F(\Delta_{UN}^\Gamma)$.*

Proof. Coassociativity follows from naturality of Δ^Γ, coassociativity of U_Γ, and naturality of η. Counitality follows from the counit laws of U_Γ and both triangle identities of $F \dashv U$. □

Theorem 3.4 (Unstable Base-Change Comonad Morphism)**.** *The natural transformation $\alpha_M := F(U_\Gamma(\eta_M)) : F(U_\Gamma(M)) \to U_\Sigma(F(M))$ makes $(F, \alpha) : (\mathcal{C}, U_\Gamma) \to (\mathcal{D}, U_\Sigma)$ a functor of comonads.*

Proof. Verified via naturality of Δ^Γ, ϵ^Γ, and η, together with the left triangle identity of $F \dashv U$. □

Theorem 3.5 (Unstable Lifted Adjunction $\alpha_* \dashv \alpha^*$)**.** *When $\mathcal{C}^{U_\Gamma}$ has equalizers, (F, α) induces an adjoint pair $\alpha_* \dashv \alpha^* : \mathcal{D}^{U_\Sigma} \to \mathcal{C}^{U_\Gamma}$, with explicit mate transformation $\beta_N = U(U_\Sigma(\varepsilon_N)) \circ \eta_{U_\Gamma(UN)}$.*

Proof. Obtained by applying Proposition 2.11 to Theorem 3.4. □

Theorem 3.6 (Lifted Counit Isomorphism $\alpha_*\alpha^* N \cong N$)**.** *Assume $\mathcal{C}^{U_\Gamma}$ has equalizers, F preserves the defining equalizer of $\alpha^* N$, and $\varepsilon : FU \xrightarrow{\sim} 1_\mathcal{D}$ is an isomorphism. Then the lifted counit $\tilde{\varepsilon}_N : \alpha_*(\alpha^* N) \to N$ is an isomorphism of U_Σ-coalgebras.*

Proof. On cofree objects, $\alpha_*(\mathrm{cofree}_{U_\Gamma}(UY)) = \mathrm{cofree}_{U_\Sigma}(Y)$ holds definitionally. Since F preserves the equalizer and ε is an isomorphism, $\alpha_*(\alpha^* N)$ is the equalizer of $(\mathrm{cofree}_{U_\Sigma}(\psi_N), \Delta^\Sigma_N)$, which is isomorphic to N by Proposition 3.1 via the lifted counit $\tilde{\varepsilon}_N$. □

Proposition 3.7 (Excess Bounds and Colimit Decomposition)**.** *The unstable excess condition $2p^n < k + (2p^n - 2)$ holds iff $k \geq 3$, and the $b(n)_*$-action excess condition $2p^n < k$ holds iff $k \geq 2p^n + 1$. In this range, left and right v_n^{-1}-actions commute with conjugation, and colimit preservation yields $U_\Gamma(K(n)_* \otimes M) \cong B(n)_* \otimes_{K(n)_*} U_\Sigma(K(n)_* \otimes M)$.*

Proof. Arithmetic bounds verified by linear arithmetic; inversion and colimit isomorphisms verified categorically. □

Proposition 3.8 (Unstable Cotensor Identification and Height-n Unit Isomorphism)**.** *When $B(n)_* \otimes_{K(n)_*} -$ preserves the canonical equalizer of U_Σ-coalgebras:*

1. *$\alpha^*(U_\Sigma(Y)) \cong U_\Gamma(U(Y))$ on cofree coalgebras.*
2. *For k-connected M with $k \geq 2p^n + 1$, the underlying object of $\alpha^*\alpha_* M$ is isomorphic to $B(n)_* \otimes_{BP_*} M$.*

3. *If M is also of height n, the lifted unit $\tilde{\sigma}_M : M \to \alpha^*\alpha_* M$ is an isomorphism of U_Γ-coalgebras.*

Proof. Proved by comparing the defining equalizer of α^* with the image of the canonical equalizer of Proposition 3.1 under $B(n)_* \otimes_{K(n)_*} -$. □

Theorem 3.9 (Unstable Change of Rings)**.** *If M is a U_Γ-comodule of height n and connectivity $k \geq 2p^n + 1$, then for any test coalgebra $P \in \mathcal{C}^{U_\Gamma}$, there is an isomorphism of coaugmented cosimplicial cobar hom-towers*

$$\mathrm{Hom}_{\mathcal{C}^{U_\Gamma}}(P, C^\bullet_{U_\Gamma}(M)) \;\cong\; \mathrm{Hom}_{\mathcal{D}^{U_\Sigma}}(\alpha_* P, C^\bullet_{U_\Sigma}(\alpha_* M)),$$

and $\alpha^*(C^\bullet_{U_\Sigma}(\alpha_* M))$ has a left-contractible hom-tower into every cofree U_Γ-coalgebra. Consequently, $\mathrm{Ext}_{U_\Gamma}(BP_*, M) \cong \mathrm{Ext}_{U_\Sigma}(K(n)_*, K(n)_* \otimes_{BP_*} M)$.

Proof. Combines the cobar isomorphism induced by the lifted unit isomorphism $\tilde{\sigma}_M : M \xrightarrow{\sim} \alpha^*\alpha_* M$ of Proposition 3.8 with the Generalized Shapiro's Lemma (Proposition 2.15). □

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